GIT-207

Practical 1

Microbiology & Immunology Department

By

Dr/ Shereen Mohamed Mohamed

Associate professor of Medical Microbiology and Immunology

Objectives:

By the end of this session, the student will be able to:

- List the main steps for stool sample examination
- list the common pathogens causing diarrhea
- Demonstrate the laboratory identification criteria of the common pathogens causing diarrhea

Laboratory examination of stools

Specimen: Stools – rectal swab

Sample processing:

1. Describe the appearance of the specimen:

A. Color of the specimen.

B. Consistency: Whether it is formed, semi-formed, unformed or fluid.

C. Presence of blood, mucus or pus.

D. Presence of worms.

2-Examine the specimen microscopically for:

A. Parasites

B. Leucocytes and red blood cells

C. Bacteria

3- Culture the specimen on:

A. MacConkey agar plates
B. Enrichment media for Salmonella and Shigella

4- Identification of isolated organism by:

Biochemical tests

A. Triple sugar iron (TSI)
B. Sugar fermentation tests

Serological tests

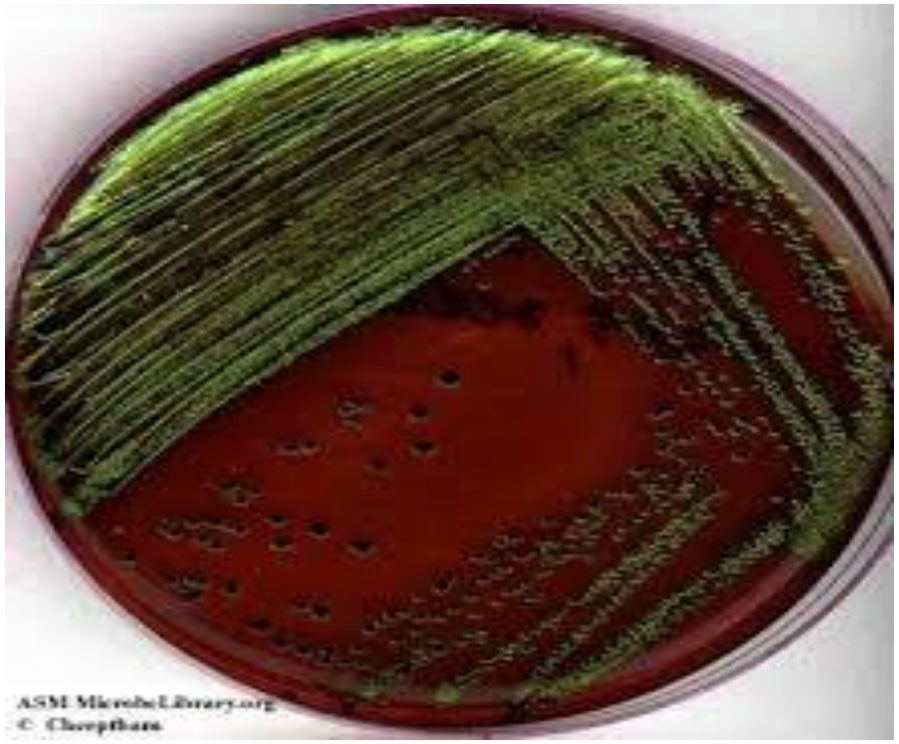
Agglutination tests with specific antisera (serotyping)

5-Detection of toxin production by :

A. Enzyme immunoassay (EIA) as ELISA
B. PCR (toxin gene)

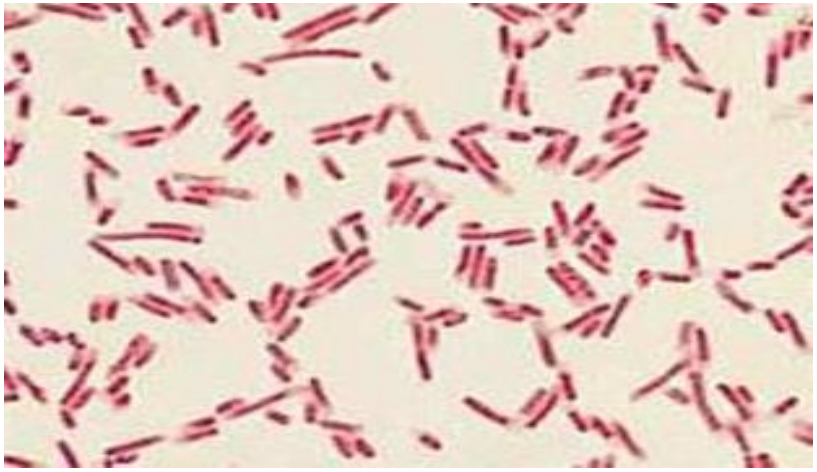
Suspected *E. coli*

On MacConkey agar	On EMB agar
Lactose fermenting organism showing rose pink colonies	dry dark colonies with green metallic sheen



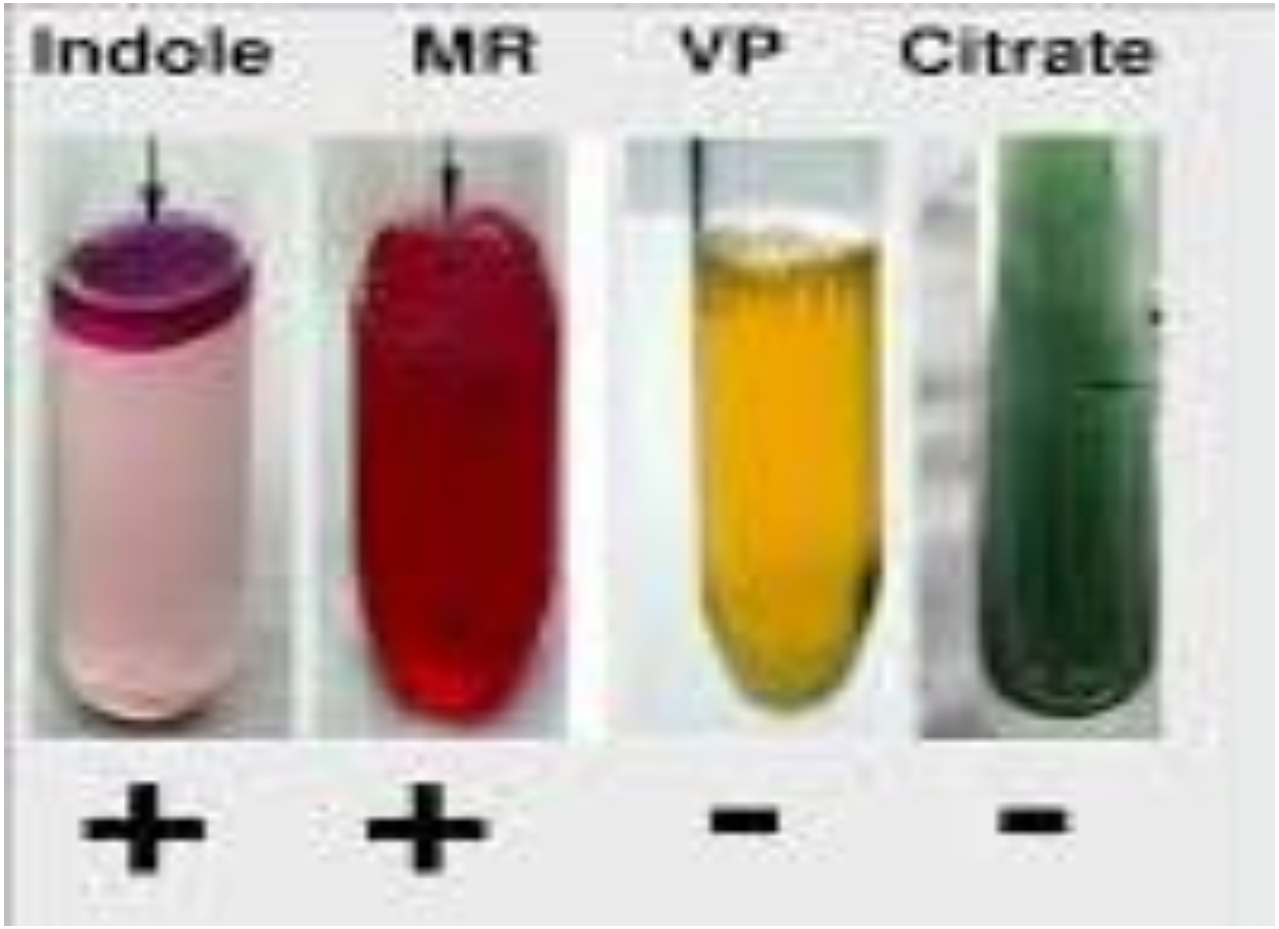
Gram-stained smear

Gram negative bacilli



Biochemical reactions

- **IMVC test (++--)**
  (indole/ Methyl red/ Vogus Proskauer/ Citrate)
- **TSI**
  Acid butt/Acid slant +gas



TSI

Suspected *Shigella*

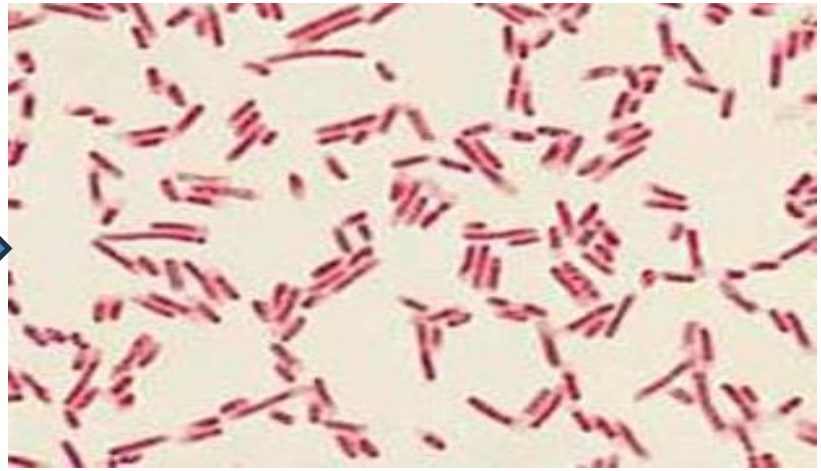
Sample is initially inoculated on selenite F broth or tetrathionate broth (enrichment) to inhibit commensal coliforms.

On MacConkey agar

non-lactose fermenting organism (pale yellow colonies)

Gram-stained smear

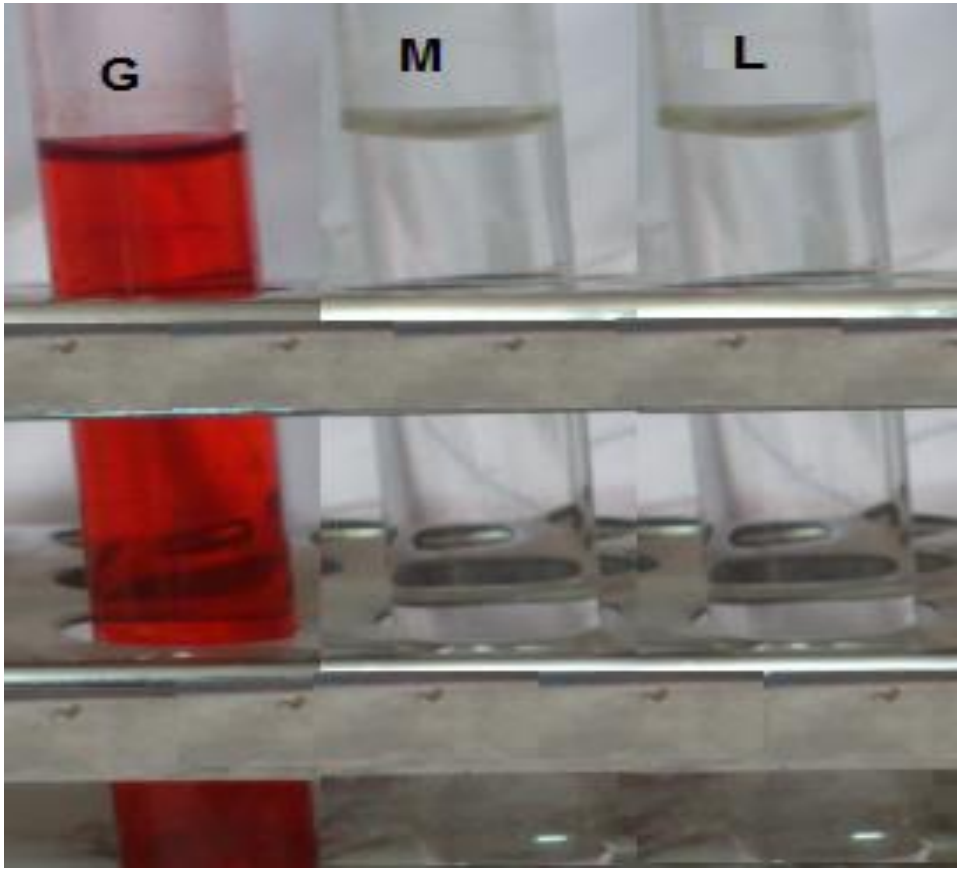
Gram negative bacilli



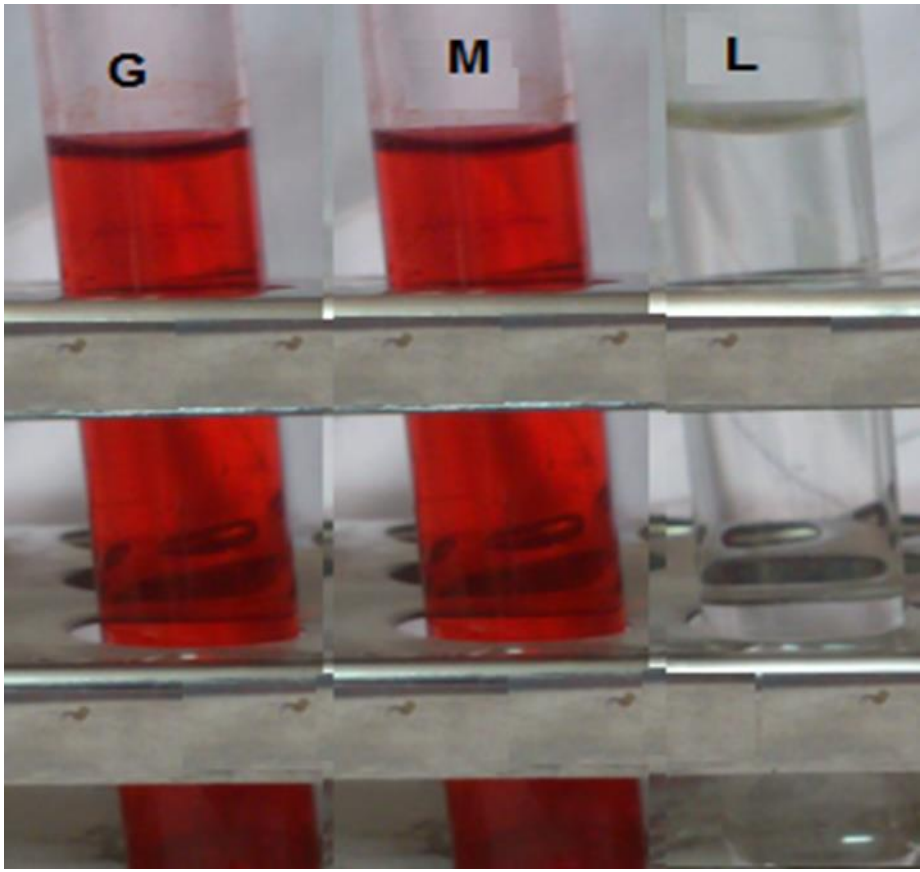
Biochemical reactions → **Sugar fermentation test**

	Glucose	Mannite	Lactose
Sh. dysenteriae	+A	-	-
Sh. Flexneri & Sh. boydii	+A	+A	-
Sh. sonnei	+A	+A	**late fermenter**

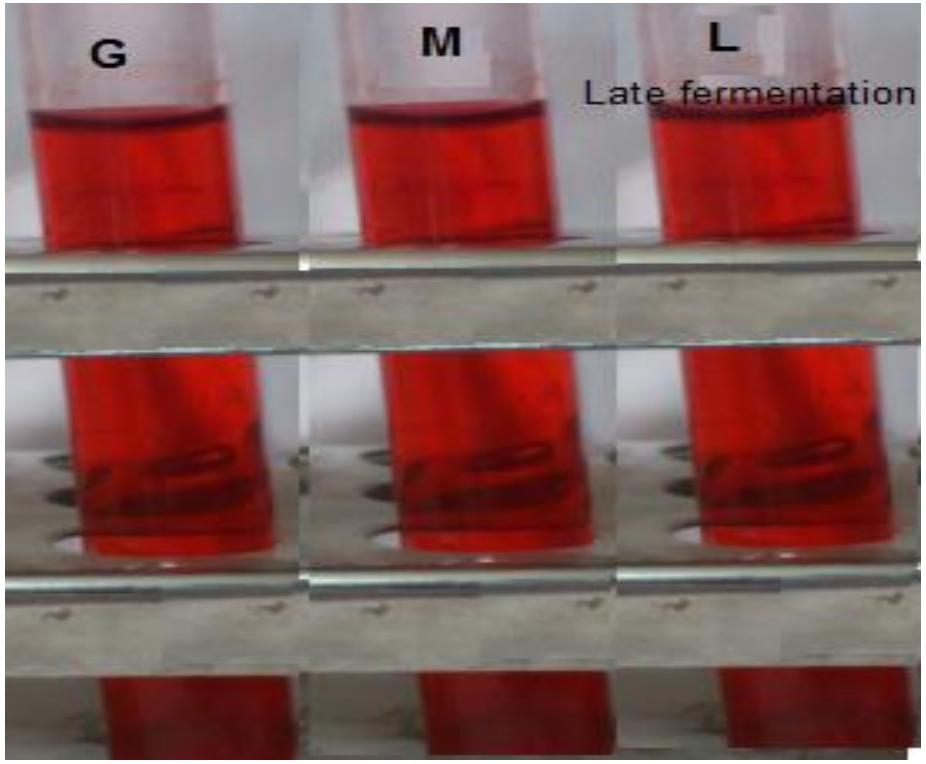
Sh. dysenteriae



Sh. flexneri& Sh. boydii



Sh. sonnei



Diagnosis of Enteric fever (Typhoid – Paratyphoid fever)

Sample:

- ***During the first week of infection*:**
- **Blood** for **blood culture**
- subculture on MacConkey agar
- ***During the second week and onward*:**
- **Urine and stool for**
  - ❑ direct isolation of the organism or
  - ❑ direct detection of salmonella antigens
- ***Serum sample for serological diagnosis***

Isolation and identification of *Salmonella*

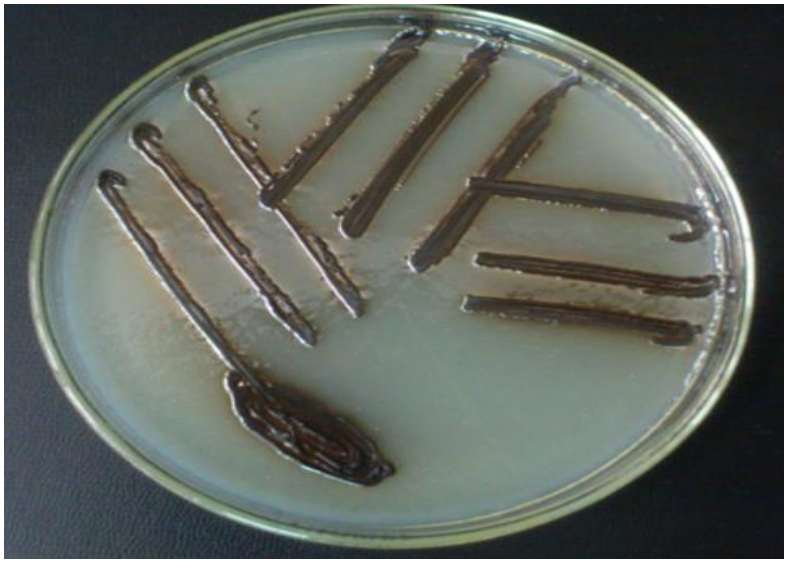
Stool sample is initially inoculated on selenite F broth or tetrathionate broth (enrichment) to inhibit commensal coliforms.

On MacConkey agar

non-lactose fermenting organism (pale yellow colonies)

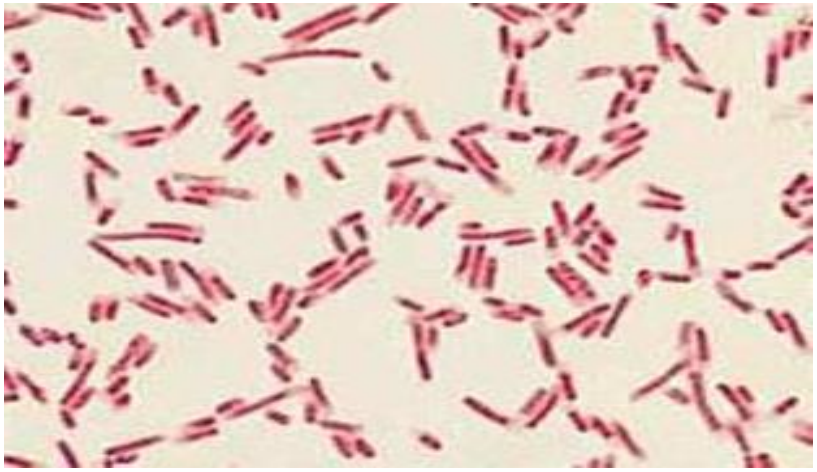
On Bismuth sulphite agar (selective)

Brown to black colonies



Gram-stained smear

Gram negative bacilli



Biochemical reactions

TSI

-Salmonella typhi produce alkaline slant/ acid butt

-Salmonella paratyphi B produce alkaline slant/ black butt (H_2S production)

Sugar fermentation test

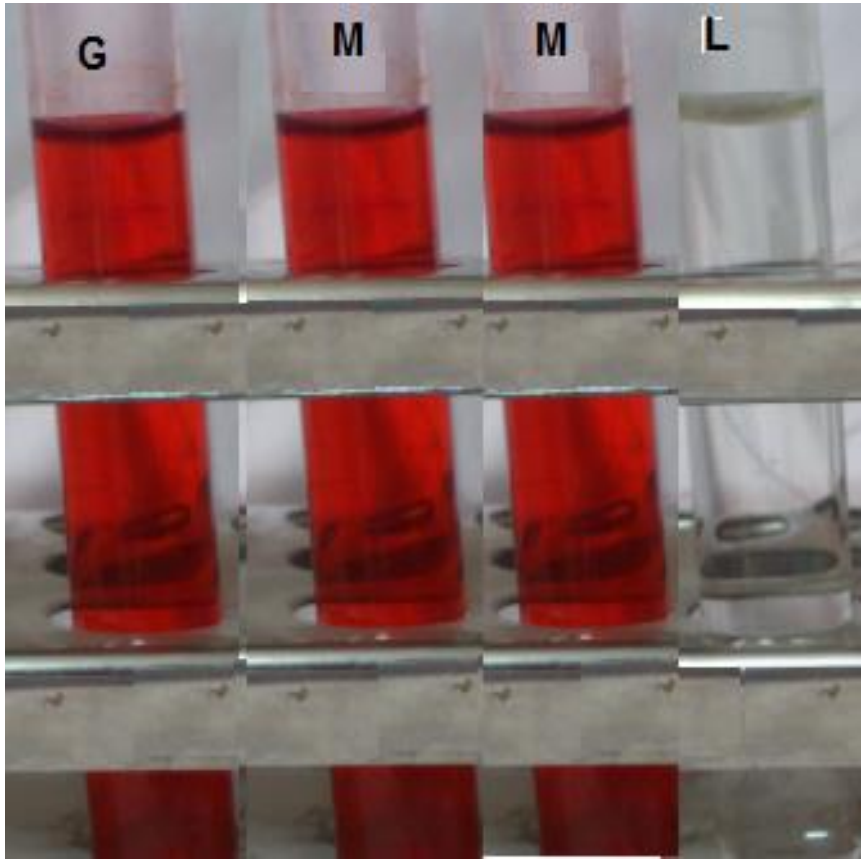
Salmonella typhi

Salmonella paratyphi B

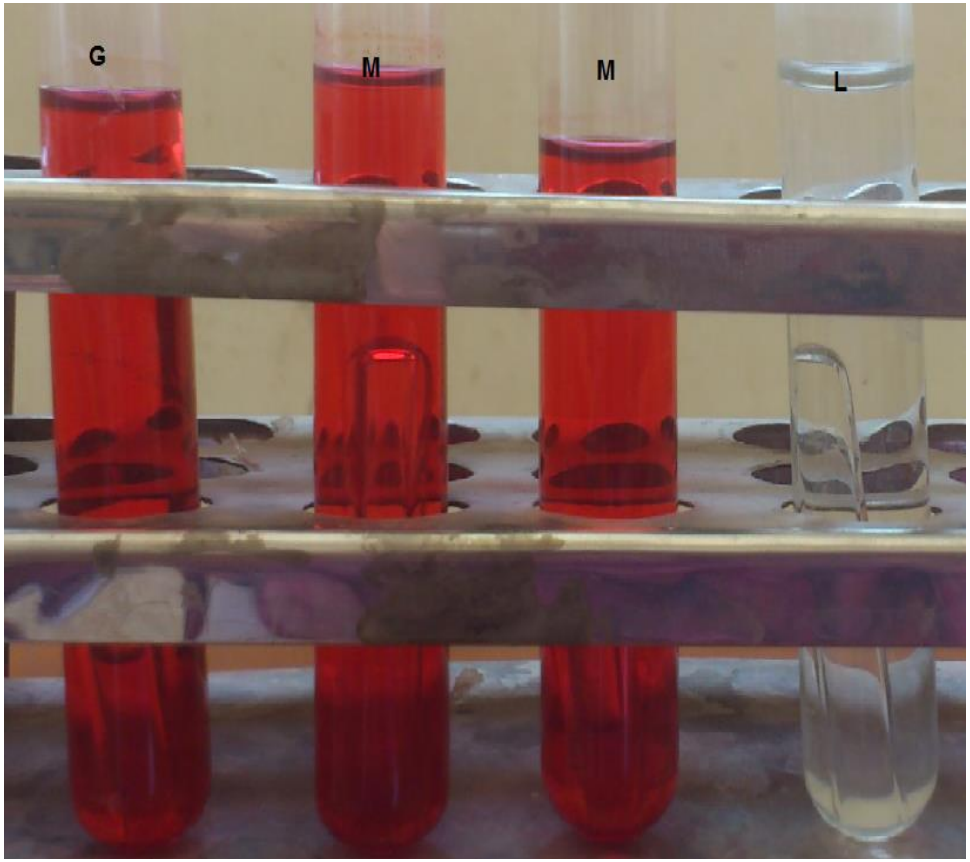
Sugar fermentation of *Salmonella*

	Glucose	*Maltose*	*Mannite*	*Lactose*	*Black butt on TSI*
S. typhi	***+A***	***+A***	***+A***	**-**	**+**
S. Paratyphi A	***+AG***	***+AG***	***+AG***	**-**	**-**
S. Paratyphi B	***+AG***	***+AG***	***+AG***	**-**	**+**

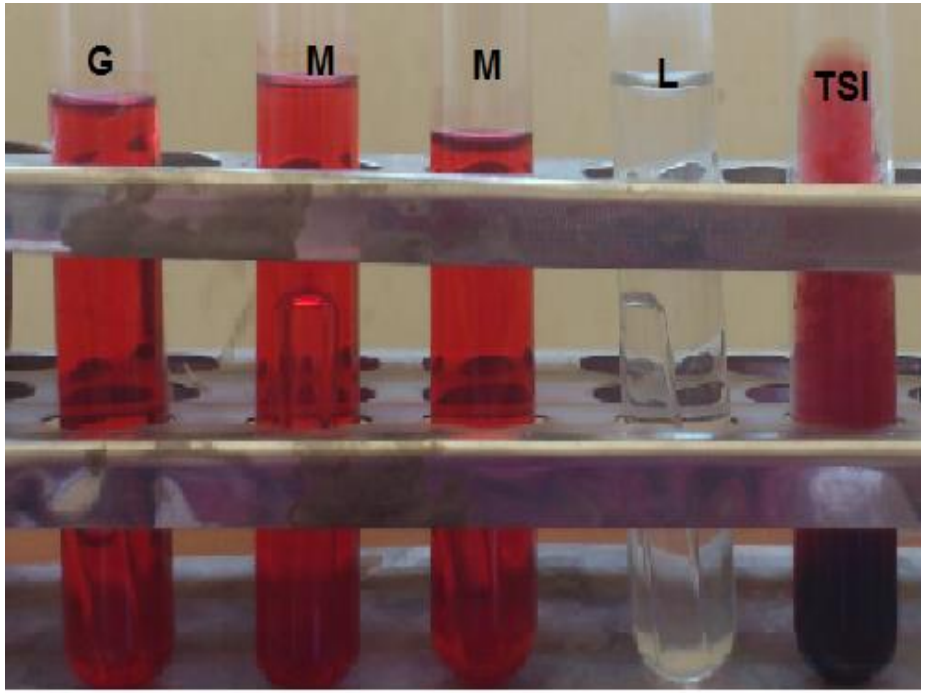
S. typhi



S. paratyphi A



S. paratyphi B



- ***Serum sample for serological diagnosis by Widal test or ELISA:***
- The antibodies to salmonella appear in the serum of the patient during the second week of illness and reach maximum titer during the 4th week.

1-Widal test

•This tests is used for detection of antibodies to antigens of *typhi*, *S. paratyphi A*, *S. paratyphi B*, *S. paratyphi C* **(rare):**

- Somatic O antigen (Antibodies to O antigen disappear faster than antibodies to H antigen).
- Flagellar H antigen
- Sometimes (Surface Vi antigen)

•Principal of the test:

If the antibody is present in patient's serum, it will react with the antigen in the reagent and gives positive agglutination.

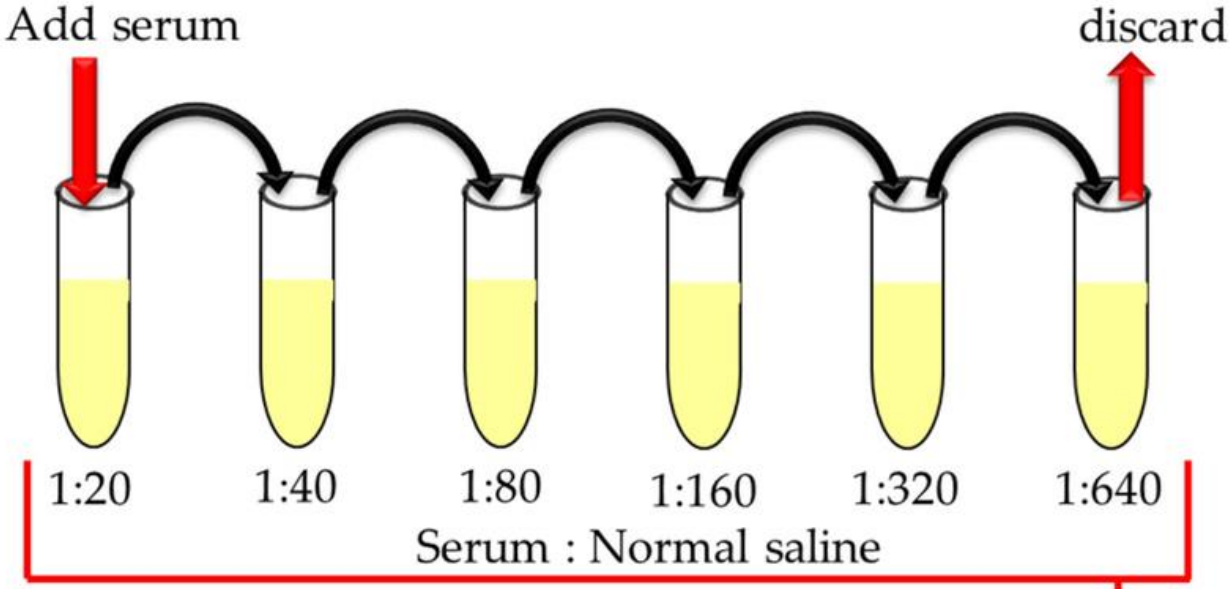
•Forms of the test:

Slide or tube agglutination (more accurate)

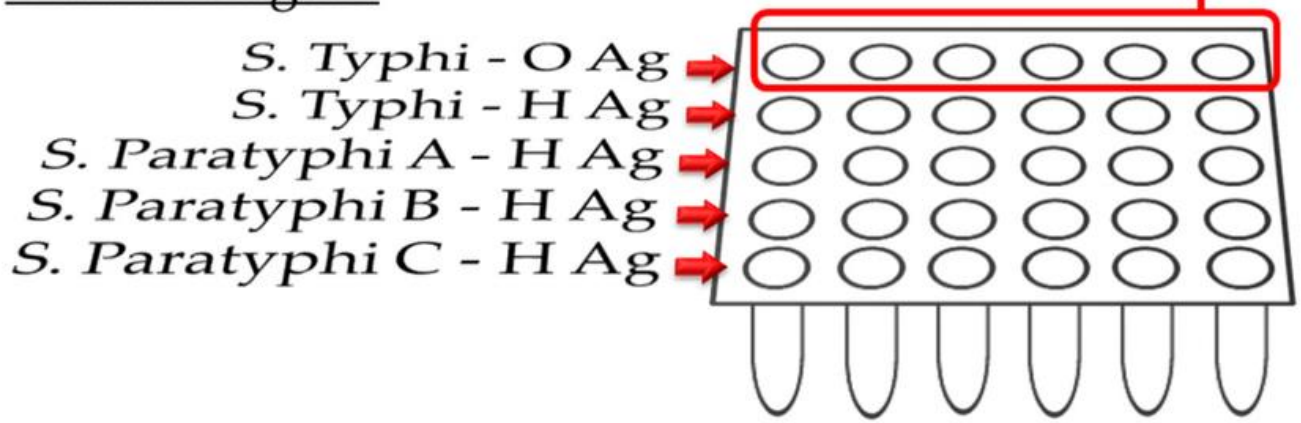
Procedure:

- Several sets of test tubes are used (one for each antigen)
- Serial dilutions of serum (with saline) (1:20, 1:40, 1:80 etc.)
- Suspension of antigens are added:
  - O antigen (common antigen)
  - H antigens of *S.typhi* , *S.paratyphi* A, *S.paratyphi* B
  - Vi ag to detect carriers.

1. Serial dilution



2. Add antigens



3. Mix properly, cover and incubate at 37° C

4. Observe for agglutination and interpret the results

- **Interpretation:**
- Titer: The highest dilution of patient serum that gives agglutination

Active enteric fever is diagnosed by:

- Rising titer in 2 serum samples 10 days apart OR
- High O antibody titer $\geq 1/160$.

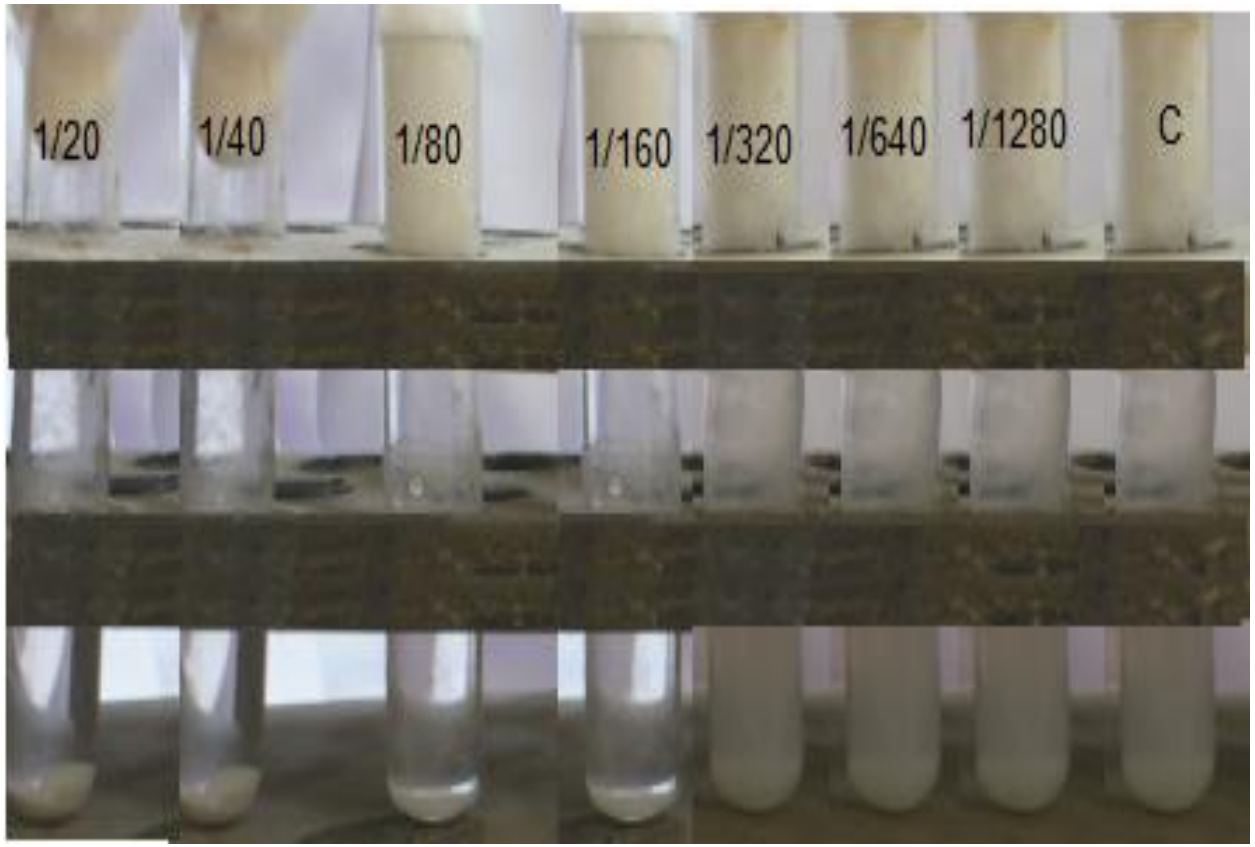
Raised H antibody helps to identify the causative pathogen of enteric fever.

Old infection or vaccination is diagnosed by:

High titer of H Antigen (alone) $\geq 1/160$.

Carrier state is diagnosed by:

Antibodies to Vi antigen.



Widal test

WIDAL TEST REPORT

Types	*1:20*	*1:40*	*1:80*	*1:160*	*1:320*	*1:640*
S. Typhi (O):	+	+	+	+	-	-
S. Typhi (H):	+	+	+	+	-	-
S.P. Typhi (AH):	+	-	-	-	-	-
S.P. Typhi (BH):	+	+	+	-	-	-

Test Result: Positive

ANY
QUESTIONS?